

Question 03

$$\text{Demand } Q_d = 100 - 2P$$

$$\text{Supply } Q_s = 4P - 20$$

$$Q_d = Q_s$$

$$100 - 2P = 4P - 20$$

$$120 = 6P$$

$$P_e = 20$$

$$Q_e = 60$$

PART A

a) tax

$$Q_d = 100 - 2(P_B)$$

$$Q_s = 4P_s - 20$$

$$P_B - P_s = 15 \Rightarrow P_{Bs} = P_B - 15 \quad \text{--- (A)}$$

$$Q_s = 4(P_B - 15) - 20$$

equilibrium point

$$100 - 2Q_d = Q_s$$

$$100 - 2P_B = 4P_B - 60 - 20$$

$$180 = 6P_B$$

$$P_B = 30$$

(A) $\Rightarrow$

$$P_s = 30 - 15$$

$$P_s = 15$$

$$Q = 100 - 2(30) = 40$$

Consumer price

Producer price = 15

equilibrium quantity = 40

b) $\text{Total tax revenue} = 40 \times 15 = \600

Consumer burden

Price increased from 20 $\rightarrow$ 30 $\rightarrow$ 10 unit

$$\text{Total tax} = 40 \times 10 = \$400 //$$

~~Consumer burden~~

20 $\rightarrow$ 15

$$\text{producer burden} = 40 \times 5$$

$$= \$200 //$$

PART B

before -

a) Demand intercept $Q_d = 100 - 2P$

maximum

$$Q = 0$$

$$P = 50$$

Supply intercept $Q_s = 4P - 20$

$$0 = 4P - 20$$

$$P = 5$$

Atlas

$$CS = \frac{1}{2} (50 - 20) \times 60 = 900$$

$$CS = 900$$

$$\text{Consumer surplus} = 900$$

$$PS = \frac{1}{2} (20 - 5) \times 60$$

$$PS = 450$$

$$\begin{aligned} \text{Total welfare} &= CS + PS \\ \text{(before)} &= 900 + 450 \end{aligned}$$

$$= 1350$$

after tax

$$CS = \frac{1}{2} (50 - 30) \times 40$$

$$CS = 400$$

$$PS = \frac{1}{2} (15 - 5) \times 40$$

$$PS = 200$$

$$\begin{aligned} \text{Total welfare} &= 400 + 200 + 600 \\ \text{(after)} & \end{aligned}$$

$$= 1200$$

$$= 1200 //$$

b)

$$\Delta WL = \text{Welfare before} - \text{Welfare after}$$

$$= 1850 - 1200$$

$$= 650$$